

StreetLight-PK

Pakistan's AI-Powered Civic Infrastructure Reporting Platform

Complete Testing Documentation | Version 1.0

Date: 2026-05-04 | Branch: test

Test Log File: documentations/test_log.md

Total Test Cases: 148 | Phases: 13

COMPONENT	TECH STACK	PORT
Backend API	FastAPI + SQLAlchemy + Supabase PostgreSQL	8000
Blockchain	Hardhat local node + Solidity contract	8545
Admin Dashboard	React (Vite) + TailwindCSS	5173
Mobile App	Flutter	N/A
AI Layers	PyTorch + OpenCV (L0-L5 pipeline)	In-process

STATUS FLOW	PENDING → VERIFIED → IN_PROGRESS → AWAITING_FEEDBACK → RESOLVED → CLOSED
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Table of Contents

1. System Overview
2. Complaint Routing & Email Accounts
3. Test Accounts
4. Startup Sequence
5. Test Phases
 - Phase 1: Health Checks
 - Phase 2: Citizen Signup & Login
 - Phase 3: AI-Powered Report Creation
 - Phase 4: Report Feed & Interactions
 - Phase 5: Community Verification
 - Phase 6: Admin Dashboard UI/UX
 - Phase 7: Status Mirroring Across All Components
 - Phase 8: Notification Pipeline
 - Phase 9: Resolution Lifecycle
 - Phase 10: Blockchain Integration
 - Phase 11: Agent Scheduler
 - Phase 12: UI/UX Functional Verification & Fixes
 - Phase 13: Edge Cases & Error Handling
6. Git Workflow
7. Test Log Template
8. Test Summary

1. System Overview

Complete Report Lifecycle

CITIZEN SUBMITS REPORT
→ AI Layer 0 (Image Validation) → Layer 1 (Classification) → Layer 2 (Fraud)
→ Layer 3 (Community Verification) → Layer 4 (Trust) → Layer 5 (Final Score)
→ Auto-Routing (GPS → City → Department → Officer)
→ Email Notification to Officer
AT EACH STAGE:
✓ Database updated
✓ Dashboard reflects change (Kanban, tables, analytics)
✓ Citizen notified (FCM push / in-app)
✓ Blockchain entry (at VERIFIED + CLOSED)
✓ Audit log entry written

2. Complaint Routing & Email Accounts

When a citizen submits a report, the system auto-routes based on **GPS city detection + AI classification**:

Photo → AI Classification (pothole/garbage)
→ GPS → City (Lahore/Faisalabad)
→ Department (LMC/LWMC/FMC/FWMC)
→ Officer assigned from routing_table
→ Email notification sent to officer's notification_email

City	Issue Type	Dept Code	Department	Officer	Notification Email
Lahore	Pothole	lmc	Lahore Metropolitan Corporation	Ahmad Raza	bitf22m006@pucit.edu.pk
Lahore	Garbage	lwmc	Lahore Waste Management Company	Sara Khan	bitf22m044@pucit.edu.pk
Faisalabad	Pothole	fmc	Faisalabad Metropolitan Corporation	Bilal Chaudhry	bitf22m011@pucit.edu.pk
Faisalabad	Garbage	fwmc	Faisalabad Waste Management Company	Ayesha Nawaz	bitf22m015@pucit.edu.pk

Note: Email notifications require SMTP_USER and SMTP_PASSWORD (Gmail App Password) in backend/.env. Without these, emails are silently skipped but routing still works.

3. Test Accounts

Password for ALL accounts:	Admin@123
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Admin Accounts (Dashboard Login)

Email	Role	Scope	Can See
super_admin@streetlight.local	super_admin	All Pakistan	Everything — all cities, departments
lahore_ca@streetlight.local	city_admin	Lahore	Only Lahore reports and officers

Email	Role	Scope	Can See
faisalabad_ca@streetlight.local	city_admin	Faisalabad	Only Faisalabad reports and officers
ahmad.raza@streetlight.local	dept_officer	Lahore / LMC	Only Lahore pothole reports
sara.khan@streetlight.local	dept_officer	Lahore / LWMC	Only Lahore garbage reports
bilal.chaudhry@streetlight.local	dept_officer	Faisalabad / FMC	Only Faisalabad pothole reports
ayesha.nawaz@streetlight.local	dept_officer	Faisalabad / FWMC	Only Faisalabad garbage reports

Citizen Accounts (Flutter App)

Any user registered via **/signup** or Flutter registration screen (default role: **citizen**).

4. Startup Sequence

Start services in this exact order. All must run simultaneously in separate terminals.

4.1 — Blockchain (Hardhat Local Node)

```
cd blockchain-layer
npm install # first time only
npx hardhat compile # compile streetLight.sol → ABI
npx hardhat node # starts at 127.0.0.1:8545

# In a second terminal – deploy:
cd blockchain-layer
npx hardhat run scripts/deploy.js --network localhost
```

Expected: ■ StreetLight deployed successfully! + Contract Address: 0x...

4.2 — Backend API

```
cd backend
pip install -r requirements.txt # first time only
uvicorn main:app --reload --host 0.0.0.0 --port 8000
```

Expected: ■ STREETLIGHT BACKEND READY + ■ Agent Scheduler Started

4.3 — Admin Dashboard

```
cd municipal-admin
npm install # first time only
# Create .env: VITE_API_BASE_URL=http://127.0.0.1:8000
npm run dev
```

Expected: Local: http://localhost:5173/

4.4 — Flutter App

```
flutter run
# Or build APK and install on phone via USB
```

Install on device via USB debug or APK

5. Test Phases

PHASE 1 — Health Checks

ID	Action	Expected Result
1.1	GET http://127.0.0.1:8000/	{"message": "StreetLight Backend Running", "version": "1.0.0"}
1.2	GET http://127.0.0.1:8000/health	{"status": "healthy"}
1.3	GET http://127.0.0.1:8000/docs	Swagger UI loads with all endpoints
1.4	Backend console — no errors	✓ Database schema ensured present
1.5	Hardhat node running	Started HTTP at http://127.0.0.1:8545
1.6	http://localhost:5173/signin	Dashboard sign-in page renders

PHASE 2 — Citizen Signup & Login

ID	Action	Expected Result
2.1	POST /signup — new citizen	{"message": "User created successfully", "user_id": }
2.2	Repeat same CNIC	400: CNIC already registered
2.3	Repeat same email	400: Email already registered
2.4	POST /login — correct credentials	{"access_token": "", "user": {"role": "citizen"}}
2.5	POST /login — wrong password	401: Invalid email or password
2.6	POST /login — non-existent email	401: Invalid email or password

PHASE 3 — AI-Powered Report Creation

ID	Action	Expected Result
3.1	Submit valid pothole photo (Lahore GPS: 31.5204, 74.3587)	success: true, issue_category: "POTHOLE", routed to Ahmad Raza
3.2	Verify routing email	bitf22m006@pucit.edu.pk received notification or log shows Email sent
3.3	Verify Cloudinary upload	report.image_url starts with https://res.cloudinary.com/
3.4	Submit valid garbage photo (Faisalabad GPS: 31.4187, 73.0791)	issue_category: "TRASH", routed to Ayesha Nawaz
3.5	Submit very blurry/dark photo	400, error_code: "VALIDATION_FAILED"
3.6	Submit non-civic image (selfie/cat/food)	400, error_code: "UNKNOWN_ISSUE_TYPE"
3.7	Submit without GPS (omit lat/long)	400: GPS coordinates are required
3.8	Submit without auth token	401 Unauthorized
3.9	Submit exact same image again	merged: true — merged into original report

ID	Action	Expected Result
3.10	Original report confirmation_count incremented	Check via GET /reports/feed

PHASE 4 — Report Feed & Interactions

ID	Action	Expected Result
4.1	GET /reports/feed?skip=0&limit;=20	Returns reports from Phase 3
4.2	Report fields present	id, image_url, issue_category, status, support_count, views
4.3	POST /reports/support	support_count +1
4.4	POST /reports/verify	verify_count +1
4.5	Category filter: GET /reports/feed?category=POTHOLE	Only pothole reports returned

PHASE 5 — Community Verification (Layer 3)

ID	Action	Expected Result
5.1	Create 2nd citizen + login	New JWT token
5.2	GET /verification/pending?lat=31.5204&lng;=74.3587 (2nd user)	Returns pending verification for report from 3.1
5.3	POST /verification/vote — {"vote": "YES"}	success: true, message: "Vote recorded"
5.4	Voter impact score increased	Check profile endpoint
5.5	GET /verification/status	Shows vote tally and status
5.6	Reporter can't verify own report	Own reports excluded from pending list

PHASE 6 — Admin Dashboard UI/UX

ID	Action	Expected Result
6.1	Login as super_admin@streetlight.local	Redirects to /dashboard
6.2	Login as lahore_ca@streetlight.local	City-scoped dashboard
6.3	Login as ahmad.raza@streetlight.local	Dept-scoped dashboard
6.4	Login with citizen account	403: Access denied
6.5	Dashboard KPI cards load	Total, Resolved, Pending counts
6.6	Recent complaints table populated	Shows reports
6.7	Navigate /complaint-management	Table loads
6.8	Click report row → detail page	Shows AI results, GPS, image
6.9	Navigate /resolution-board	Kanban columns render
6.10	Drag card VERIFIED → IN_PROGRESS	Card moves, API fires
6.11	Refresh — card stays in position	Persisted correctly
6.12	Navigate /hotspot-map	Map renders with pins
6.13	Click pin → detail panel	Report info shown
6.14	Navigate /analytics	Charts and KPIs render
6.15	Change time range (7/30/90 days)	Charts update
6.16	Navigate /organization (super_admin)	Routing table + users tabs
6.17	/organization as dept_officer	Redirected to /dashboard
6.18	Navigate /audit-log	Audit entries load
6.19	Navigate /transparency	KPI data loads
6.20	Sign out → redirects to /signin	Token cleared

PHASE 7 ■ — Status Mirroring Across All Components

Critical: When status changes in ONE component, ALL components must reflect it.

7A — VERIFIED → IN_PROGRESS (Admin Dashboard)

ID	Action	Check Component	Expected Result
7.1	Drag card to IN_PROGRESS in Kanban	Dashboard	Card in IN_PROGRESS column
7.2		Database (Supabase)	kanban_stage = IN_PROGRESS, status = IN_PROGRESS
7.3		Flutter App	Report shows status: IN_PROGRESS
7.4		Audit Log	Log entry: stage change VERIFIED → IN_PROGRESS

7B — IN_PROGRESS → AWAITING_FEEDBACK (After-image Upload)

ID	Action	Check Component	Expected Result
7.5	Officer uploads after-image via complaint detail	Dashboard	After-image visible, stage = AWAITING_FEEDBACK
7.6		Database	after_image_url populated, kanban_stage = AWAITING_FEEDBACK
7.7		Flutter App	Citizen sees notification to confirm resolution
7.8		Audit Log	Log entry with after-image upload + stage change

7C — AWAITING_FEEDBACK → RESOLVED → CLOSED (Citizen Confirms)

ID	Action	Check Component	Expected Result
7.9	Citizen confirms resolution in Flutter app	Database	citizen_response = CONFIRMED, status = CLOSED, closed_at set
7.10		Dashboard (Kanban)	Card in CLOSED column
7.11		Dashboard (Analytics)	Resolved count incremented
7.12		Blockchain	markResolved() called — TX hash stored
7.13		Hardhat Console	Transaction confirmed
7.14		Flutter App	Report shows CLOSED status
7.15		Audit Log	2 entries: RESOLVED + CLOSED with blockchain TX

7D — Citizen REJECTS Resolution

ID	Action	Check Component	Expected Result
7.16	Citizen rejects resolution in Flutter	Database	citizen_response = REJECTED, kanban_stage = IN_PROGRESS
7.17		Dashboard (Kanban)	Card moved back to IN_PROGRESS
7.18		Officer (push notification)	Gets "Citizen rejected resolution" push

ID	Action	Check Component	Expected Result
7.19		After-image cleared	after_image_url = NULL
7.20		Audit Log	Log entry: AWAITING_FEEDBACK → IN_PROGRESS with rejection note

7E — RESOLVED Without After-Image (Agent Catches)

ID	Action	Check Component	Expected Result
7.21	Try to resolve without after-image	Agent	Reverts to IN_PROGRESS automatically
7.22		Officer	Gets "After-image required" push
7.23		Database	kanban_stage = IN_PROGRESS
7.24		Audit Log	RESOLVED attempted but no after-image uploaded

PHASE 8 ■ — Notification Pipeline

Critical: Citizen must be notified at each meaningful status change. Verify backend logs for FCM=sent/skipped and Email sent entries.

ID	Status Change	Notification Type	Recipient	Expected Message
8.1	Report created + routed	In-app + push (FCM)	Citizen (reporter)	Your report has been routed to [DEPT] in [CITY]
8.2	Report created + routed	Email	Officer (assigned)	Email with report details to notification_email
8.3	Officer uploads after-image	In-app + push (FCM)	Citizen	Your issue has been fixed! Please confirm
8.4	Citizen rejects resolution	Push (FCM)	Officer	Citizen rejected resolution
8.5	Resolution finalized (CLOSED)	Push (FCM)	Citizen	Complaint Closed — Thank you!
8.6	After-image fails verification	Push (FCM)	Officer	After-image rejected — Please re-upload
8.7	Agent auto-closes (7 days timeout)	Push (FCM)	Citizen	Complaint Closed
8.8	Report verified by community votes	In-app	Citizen	Status updated to VERIFIED

PHASE 9 — Resolution Lifecycle (Full End-to-End)

ID	Action	Expected Result
9.1	Create report via Flutter app (Lahore pothole)	Report created, routed to Ahmad Raza
9.2	Login as ahmad.raza@streetlight.local in dashboard	See the report in complaint list
9.3	Move report to IN_PROGRESS in Kanban	Stage updated in DB + dashboard
9.4	Upload after-image via complaint detail page	after_image_url populated, stage → AWAITING_FEEDBACK
9.5	Citizen gets FCM/in-app notification	Your issue has been fixed! Please confirm
9.6	Citizen opens Flutter app → Resolution screen	Sees before/after images
9.7	Citizen taps "Confirm"	POST /reports/resolution/confirm → CONFIRMED
9.8	Report → RESOLVED → blockchain → CLOSED	Full finalization chain
9.9	Dashboard Kanban shows card in CLOSED	Mirrored
9.10	Database: status=CLOSED, closed_at set, blockchain_resolve_tx populated	All fields correct
9.11	Analytics: resolved count +1	Dashboard analytics updated
9.12	Citizen gets final "Complaint Closed" notification	Notification received

PHASE 10 — Blockchain Integration

ID	Action	Expected Result
10.1	Report reaches VERIFIED (score >= 85 auto)	registerComplaint() called on-chain
10.2	Backend logs	BLOCKCHAIN: Registering complaint # + TX hash
10.3	Hardhat console	Transaction confirmed in block
10.4	Read on-chain proof: getComplaint()	Returns imageHash, aiScore, finalScore, status: VERIFIED
10.5	getStats() on contract	total >= 1, pending >= 1
10.6	Run python script/test_blockchain.py	Registers complaint #999 + reads proof
10.7	After CLOSED: markResolved() on-chain	blockchain_resolve_tx stored in DB
10.8	On-chain status changes to RESOLVED	getComplaint() shows status: RESOLVED

PHASE 11 — Agent Scheduler

ID	Action	Expected Result
11.1	GET /test-agent (super_admin JWT)	{"message": "Agent cycle completed"}
11.2	Backend logs	Agent cycle started → Found X active reports → completed
11.3	Reports in PENDING with high scores	May auto-transition to VERIFIED
11.4	Wait 15 min after startup	Agent runs automatically
11.5	GET /test-agent (non-super_admin)	403: Only super_admin may trigger
11.6	Auto-close test: AWAITING_FEEDBACK > 7 days	check_auto_close finalizes the report

PHASE 12 ■ — UI/UX Functional Verification & Fixes

Goal: Verify every interactive UI element works. If something is broken (e.g. search bar), FIX it and commit.

Dashboard UI Elements

ID	Element	Page / Screen	Test Action	Expected Result
12.1	Search bar (top nav)	All pages	Type query, press Enter	Filters reports by search term
12.2	Search in Complaint Management	/complaint-management	Type report ID or title	Table filters to matching reports
12.3	Search in Resolution Board	/resolution-board	Type query	Kanban filters cards
12.4	Search in User Roles	/organization?tab=users	Type officer name/email	List filters
12.5	Stage filter dropdown	/complaint-management	Select a stage	Only reports in that stage shown
12.6	Date range filter	/complaint-management	Select date range	Reports filtered by date
12.7	Clear filters button	/complaint-management	Click "Clear"	All filters reset
12.8	Category filter on map	/hotspot-map	Select POTHOLE only	Only pothole pins visible
12.9	Export CSV button	/analytics	Click export	CSV file downloads
12.10	PDF report modal	/analytics	Click generate report	PDF opens/downloads
12.11	Kanban drag-and-drop	/resolution-board	Drag card between columns	Card moves + API call fires
12.12	Pagination	/complaint-management	Click next page	New page of results loads
12.13	Responsive layout	All pages	Resize browser window	Layout adapts without breaking
12.14	Sign out button	Top nav	Click sign out	Token cleared, redirect to /signin
12.15	Profile page	/my-profile	Load page	Shows current admin user info

Flutter App UI Elements

ID	Element	Page / Screen	Test Action	Expected Result
12.1 6	Splash screen	Launch	Open app	Logo + loading animation
12.1 7	Login form validation	Login	Submit empty fields	Validation errors shown
12.1 8	Registration form	Registration	Fill all fields + submit	Account created
12.1 9	Report submission	Report Issue	Take photo, fill fields, submit	Report created with AI result
12.2 0	Feed infinite scroll	Home	Scroll down	More reports load
12.2 1	Support/Verify buttons	Report detail	Tap support	Count increments
12.2 2	Verification screen	Verification tab	View nearby pending requests	List loads within 2km
12.2 3	Resolution confirmation	Resolution screen	Confirm/Reject resolution	Status updates
12.2 4	Notification bell badge	Home	New notifications exist	Badge shows unread count
12.2 5	Profile screen	Profile tab	Load screen	Stats, impact score visible

PHASE 13 — Edge Cases & Error Handling

ID	Scenario	Expected Result
13.1	Submit report with no auth token	401 Unauthorized
13.2	Access admin API with citizen JWT	403 Forbidden
13.3	Admin login with expired JWT	401 on API call
13.4	GPS outside Pakistan (e.g. 0, 0)	Report saves but routing fails: kanban_stage = NEW
13.5	10+ reports in 1 hour (same user)	is_flagged_for_spam: true, penalty applied
13.6	BLOCKCHAIN_ENABLED=false	Backend runs; blockchain calls return skipped: true
13.7	AI model file missing	Fallback mode: final_score = 50, agent_decision = REVIEW
13.8	Invalid Kanban stage in PATCH	400: Bad Request
13.9	dept_officer views other city's reports	Empty list (scope filter)
13.10	Report in Islamabad (no routing entry)	Saved but not routed; kanban_stage = NEW
13.11	Double-vote on same verification	Error: already voted
13.12	Confirm resolution on non-AWAITING report	Error: "Report not awaiting feedback"

6. Git Workflow

```
# Create and switch to test branch
git checkout -b test

# After each test phase or fix:
git add .
git commit -m "test(phase-X): "

# Example commits:
# git commit -m "test(phase-1): verify health checks - all passed"
# git commit -m "fix(dashboard): search bar not filtering in complaint management"
# git commit -m "test(phase-7): status mirroring across all components verified"
# git commit -m "docs(test-log): add results for phases 1-6"

# Push to remote
git push origin test
```

Commit Convention

test(phase-X):	Test result documentation
fix(component):	Bug fix found during testing
docs(test-log):	Test log updates

7. Test Log Template

Every test case must be logged in **documentations/test_log.md** using this format:

Test ID	X.Y
Date	YYYY-MM-DD HH:MM
Tester	[Name]
Component	Backend / Dashboard / Flutter / Blockchain
Action Taken	Exact steps performed
Expected Result	What should happen
Actual Result	What actually happened
Status	PASS / FAIL / PARTIAL
Fix Applied	(if FAIL) What was changed to fix it
Commit	(if fix) Commit hash + message
Screenshots	(if applicable) Path to screenshot
Notes	Any additional observations

8. Test Summary

Phase	Description	Total Tests	Passed	Failed	Fixed	Notes
1	Health Checks	6				
2	Citizen Signup & Login	6				
3	AI Report Creation	10				
4	Feed & Interactions	5				
5	Community Verification	6				
6	Admin Dashboard UI/UX	20				
7	Status Mirroring	24				
8	Notification Pipeline	8				
9	Resolution Lifecycle	12				
10	Blockchain Integration	8				
11	Agent Scheduler	6				
12	UI/UX Functional & Fixes	25				
13	Edge Cases	12				
TOTAL		148				

Fill in Passed, Failed, and Fixed columns as testing progresses. Log all results in documentations/test_log.md per the template in Section 7.